Business Knowledge | Supply Chain

1. Orders vs. Order Lines

Understanding the hierarchy of a transaction is fundamental to supply chain analytics:

- **Order**: A unique request placed by a customer on a specific date, identified by a single **Order ID**.
- **Order Line**: Each individual item type requested within that single order. An order can contain multiple lines if a customer buys different products (e.g., notebooks and pens).
- **Key Distinction**: Total order lines will always be equal to or greater than the total number of unique orders.

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2. Supply Chain Performance Metrics

These KPIs help the supply planning and sales teams evaluate how well the inventory is meeting customer demand.

Line Fill Rate (LFR)

This measures the reliability of fulfilling the variety of items requested, regardless of delivery speed.

- **Definition**: The ratio of order lines fulfilled in full to the total number of lines ordered.
- **Formula**:

$$\text{Line Fill Rate} = \frac{\text{Number of Lines Fulfilled in Full}}{\text{Total Lines Ordered}}$$

- **Goal**: Helps the planning team understand if they are stocking the right mix of products.

Volume Fill Rate (VFR)

Also known as **Case Fill Rate**, this focuses on the sheer quantity of products shipped.

- **Definition**: The ratio of the total quantity of units shipped to the total quantity of units ordered over a specific period.
- **Formula**:

$$\text{Volume Fill Rate} = \frac{\text{Total Quantity Shipped}}{\text{Total Quantity Ordered}}$$

- **Goal**: Frequently used by sales teams during negotiations to show overall fulfillment volume to a customer.

Following the same structured format as the Fill Rate metrics, here is the breakdown of the high-stakes, order-level KPIs mentioned in the project:

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3. Order-Level Performance Metrics

While Fill Rates are calculated at the line or volume level, the following metrics are measured at the **Order Level**. An entire order is marked as a "fail" if even a single line item within it does not meet the criteria.

On Time Delivery %

This tracks the punctuality of the logistics chain.

- **Definition**: Determines if an order was delivered by the specific date agreed upon with the customer.
- **Rule**: An order is considered "On Time" only if **all** line items within that order are delivered on time.
- **Stakeholders**: Primarily used by the **Warehouse & Distribution** teams to measure shipping efficiency.

In Full Delivery %

This tracks the accuracy of fulfillment quantities.

- **Definition**: Determines if the total quantity requested for an order was delivered as specified.
- **Rule**: An order is "In Full" only if **every** line item within that order is delivered in the exact quantity requested.
- **Stakeholders**: Crucial for the **Supply Planning** team to monitor inventory availability.

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4. The "Perfect Order" Metric: OTIF

On Time In Full (OTIF) is the most rigorous standard in supply chain analytics.

- **Definition**: Measures if an order was delivered **BOTH** In Full and On Time.
- **The "Hard" Metric**: It is considered a "harsh" metric because if 199 items in a 200-line order are perfect, but one item is missing or late, the entire order fails the OTIF test.

- **Reliability**: From a customer’s perspective, OTIF is the ultimate measure of a supplier's reliability and is often used to draft Service Level Agreements (SLAs).
- **Stakeholders**: Monitored by senior leadership, including **Supply Chain VPs and Directors**, as it reflects the performance of all sub-functions.